

EdgeTRM Model Evaluation & Compression Report

EdgeTRM Model Evaluation & Compression Report

This report consolidates the performance scorecards, parameters, footprints, and visual plots extracted from the three core model variants: 1. **ARC-Prize 2024 (Baseline)** 2. **Maze (Hard Version)** 3. **Sudoku (Extreme Version)**

ARC 2024 Evaluation Outputs

MUST RUN FIRST — fix duplicate trm.py on Modal volume

Cell 9: Sparsity Audit

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

4.1 ARC DataLoader from pre-built .npy files

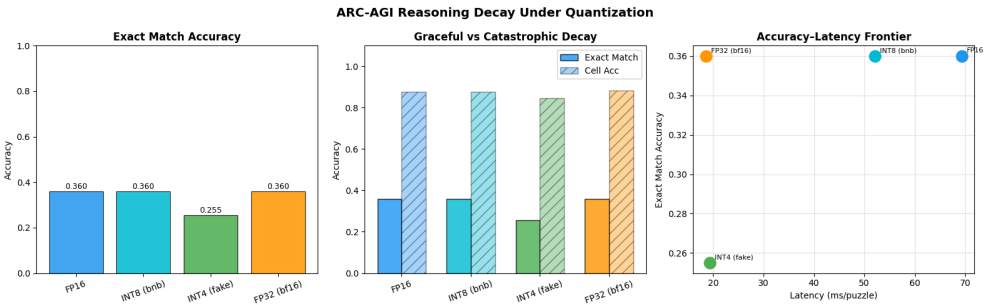
4.3 Baseline: evaluate the loaded FP32 model

4.4 Quantization helpers

4.7 Instantiate all model variants

5.1 Evaluate all quantization levels

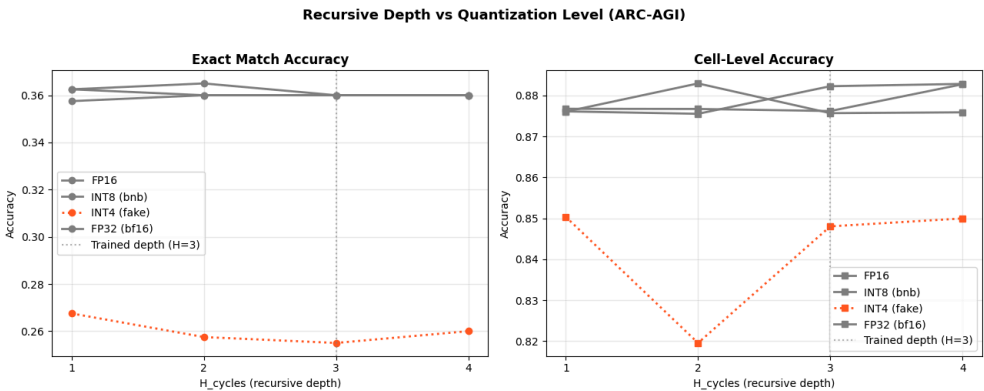
5.2 Visualise reasoning decay



Plot from cell 23

6.1 Recursive depth sweep at inference time

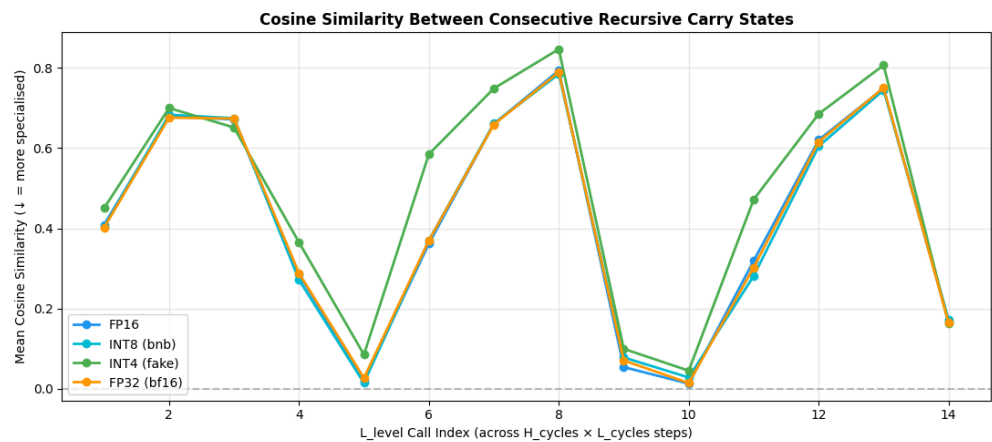
6.2 Plot depth × quantization grid



Plot from cell 26

7.1 Carry-state similarity hooks

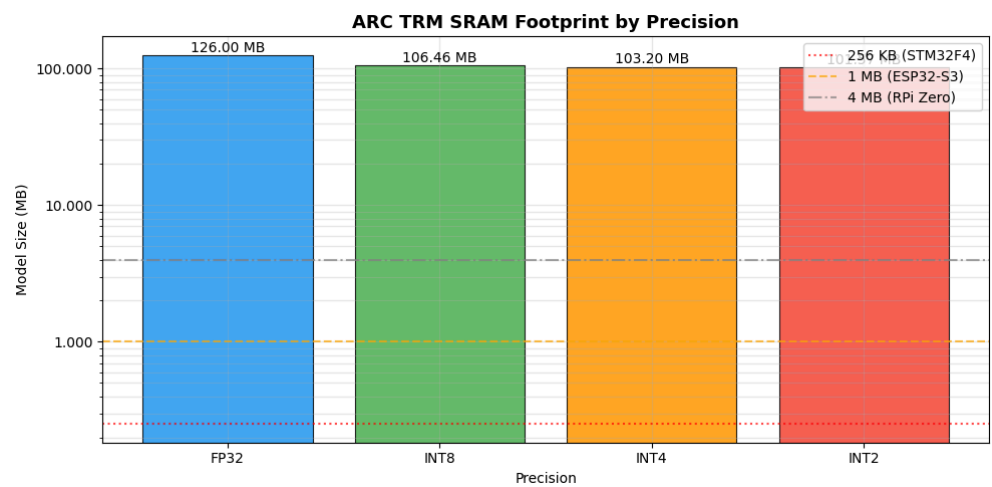
7.2 Compare carry similarity across quantization variants



Plot from cell 29

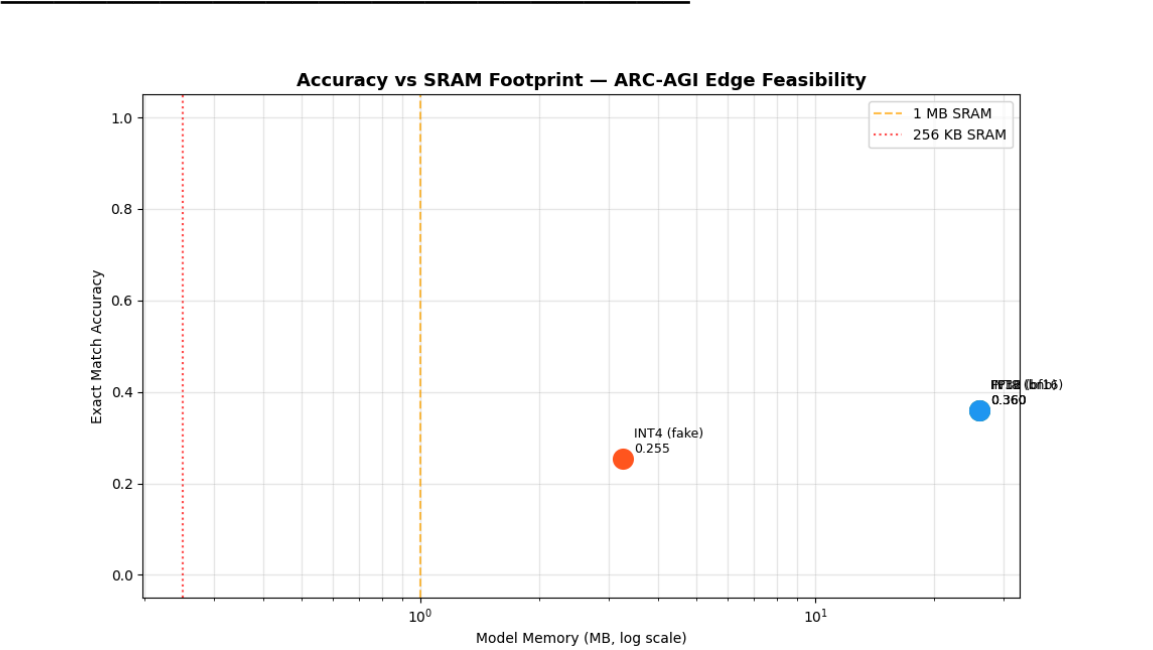
8.1 Comprehensive model footprint analysis

8.2 SRAM footprint visualisation



Plot from cell 33

8.3 Accuracy vs SRAM: the key publishable plot



Plot from cell 34

9.1 Inspect puzzle_emb shape & byte count

9.2 Option A — INT8 quantize the embedding buffer

9.3 Option B — SVD Low-Rank Puzzle Embedding

9.4 Option D — Single-Puzzle Inference (only load 1 row)

9.5 Footprint summary across embedding strategies

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

11.1 Per-channel calibration statistics

11.2 Apply calibrated INT4 and evaluate

12.1 QAT fine-tuning loop

13.1 Magnitude-based row pruning

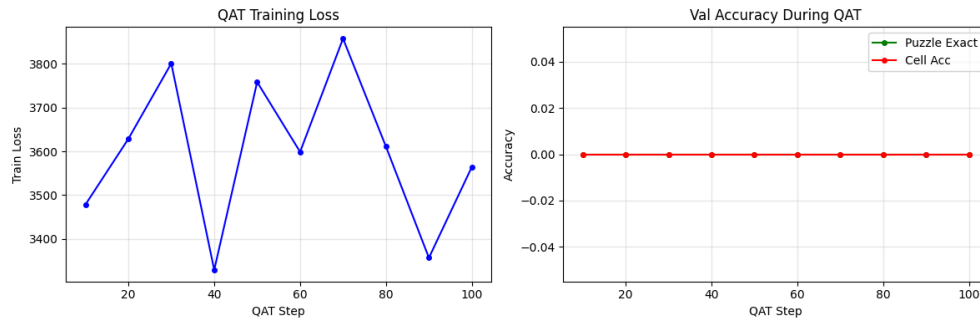
14.1 TorchScript trace of single TRM backbone step

14.2 Latency benchmark: traced model vs eager

15.1 Build train / val loaders

15.2 QAT loop with val-set evaluation every 10 steps

15.3 Plot train loss + val accuracy curve



Plot from cell 62

16.1 Build fused INT8 + SinglePuzzle model

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

16.3 Save fused model (state dict + embedding table)

17.1 CPU-only latency benchmark

17.2 Peak SRAM usage during inference (tracemalloc)

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 187504.0 MFL0Ps
 Full inference (16 steps) : 3000.06 GFL0Ps

Power estimate (Cortex-M55 @ ~1 GFL0PS/s, ~1 mW/GFL0P):
 FLOPs : 3000.064 GFL0Ps/puzzle

Est. energy : ~3000.06 mJ/puzzle

Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

=== Final Deployment Scorecard ===

	Backbone MB	Total deploy MB	Puzzle Exact	Cell
Acc Fits 4MB Fits 8MB				
Config				
FP32 (baseline)	26.05	86.46	0.0000	
0.0000 x x				
INT8 (bnb)	6.51	66.92	0.0246	
0.2003 x x				
INT4 (calibrated)	3.26	3.26	0.0148	
0.1902 ✓ ✓				
INT8 + Single-Puzzle ★	6.51	6.51	0.0000	
0.0000 x ✓				

★ = recommended target for 8MB SRAM devices

Utility: fix H_init / L_init device mismatch (permanent patch)

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

3 (Def)	Exact	0.2850	0.3600	0.3575	0.3600
0.3625					
	Cell	0.8762	0.8838	0.8765	0.8763
0.8755					
	Latency	7.5 ms	14.4 ms	28.1 ms	55.7 ms
111.0 ms					
	GFLOPs	187.5	375.0	750.0	1500.0
3000.1					

1	Exact	0.0025	0.0750	0.3450	0.3550
0.3575					
	Cell	0.5410	0.7953	0.8825	0.8831
0.8773					
	Latency	3.0 ms	5.3 ms	9.9 ms	19.2 ms
37.6 ms					
	GFLOPs	62.5	125.0	250.0	500.0
1000.0					
=====					
=====					

VARIANT: INT8 (bnb)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.2850	0.3575	0.3575	0.3600
0.3600	Cell	0.8760	0.8837	0.8827	0.8842
0.8670	Latency	5.8 ms	11.0 ms	21.3 ms	41.9 ms
83.2 ms	GFL0Ps	187.5	375.0	750.0	1500.0
3000.1					
1	Exact	0.0025	0.0750	0.3425	0.3525
0.3625	Cell	0.5415	0.7953	0.8827	0.8835
0.8767	Latency	2.3 ms	4.1 ms	7.6 ms	14.6 ms
28.6 ms	GFL0Ps	62.5	125.0	250.0	500.0
1000.0					

VARIANT: INT4 (fake)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.1575	0.2700	0.2675	0.2625
0.2575	Cell	0.8372	0.8626	0.8473	0.8430
0.8322	Latency	2.6 ms	4.4 ms	8.1 ms	15.5 ms
30.3 ms	GFL0Ps	187.5	375.0	750.0	1500.0
3000.1					
1	Exact	0.0000	0.0225	0.2525	0.2650
0.2600	Cell	0.3954	0.7148	0.8582	0.8494

0.8415					
	Latency	1.3 ms	1.9 ms	3.2 ms	5.7 ms
10.7 ms					
	GFL0Ps	62.5	125.0	250.0	500.0
1000.0					
=====					

=====					
VARIANT: FP32 (bf16)		(device: cuda)			
=====					
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					

3 (Def)	Exact	0.2850	0.3600	0.3575	0.3575
0.3600					
◀	Cell	0.8762	0.8839	0.8764	0.8760 ▶
0.8757					
	Latency	7.5 ms	14.3 ms	28.2 ms	55.9 ms
111.2 ms					
	GFL0Ps	187.5	375.0	750.0	1500.0
3000.1					

1	Exact	0.0025	0.0750	0.3425	0.3575
0.3550					
	Cell	0.5410	0.7954	0.8825	0.8832
0.8762					
	Latency	3.0 ms	5.2 ms	9.9 ms	19.2 ms
37.6 ms					
	GFL0Ps	62.5	125.0	250.0	500.0
1000.0					
=====					
◀					▶

19.1 Zero-pad truncation at various context lengths

900	100%	~187 GFL0Ps	0.3600	0.8755
450	50%	~47 GFL0Ps	0.2650	0.2500
225	25%	~12 GFL0Ps	0.1075	0.0492
128	14%	~4 GFL0Ps	0.0825	0.0595

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

21.1 Instantiate student TRM and train with KL distillation

22.1 Monkey-patch softmax attention → linear attention

22.2 Retrain the linear attention model (distillation)

Maze Hard Evaluation Outputs

MUST RUN FIRST — fix duplicate trm.py on Modal volume

Cell 9: Sparsity Audit

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

4.1 Maze-Hard DataLoader from pre-built .npz files

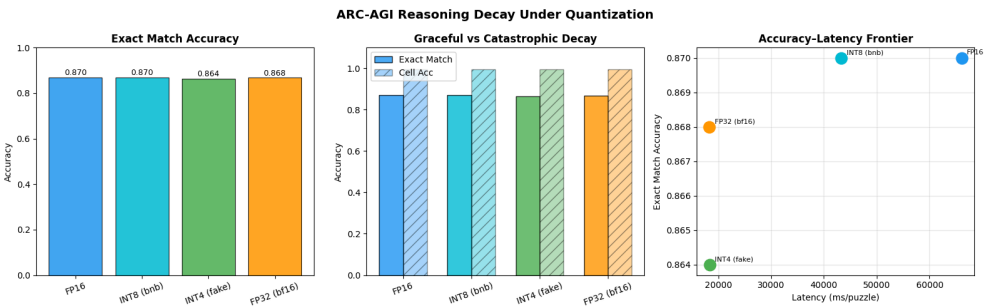
4.3 Baseline: evaluate the loaded FP32 model

4.4 Quantization helpers

4.7 Instantiate all model variants

5.1 Evaluate all quantization levels

5.2 Visualise reasoning decay



Plot from cell 31

6.1 Recursive depth sweep at inference time

Quant GFL0Ps	H	n_sup	Pass@1	Pass@2	Cell Acc	Latency

FP16	1	1	0.0000	0.0000	0.4362	2.68
62.5						
FP16	1	2	0.1380	0.1380	0.9808	4.86
125.0						
FP16	1	4	0.8440	0.8440	0.9949	9.25
250.0						
FP16	1	6	0.8670	0.8670	0.9953	13.64
375.0						
FP16	1	8	0.8700	0.8700	0.9953	18.04
500.0						
FP16	1	10	0.8700	0.8700	0.9953	22.42
625.0						
FP16	2	1	0.1380	0.1380	0.9808	4.86
125.0						
FP16	2	2	0.8440	0.8440	0.9949	9.24
250.0						
FP16	2	4	0.8700	0.8700	0.9953	18.00
500.0						
FP16	2	6	0.8700	0.8700	0.9953	26.79
750.0						
FP16	2	8	0.8700	0.8700	0.9953	35.51
1000.0						
FP16	2	10	0.8700	0.8700	0.9953	44.21
1250.0						
FP16	3	1	0.7850	0.7850	0.9940	7.05
187.5						
FP16	3	2	0.8670	0.8670	0.9953	13.61
375.0						
FP16	3	4	0.8700	0.8700	0.9953	26.72
750.0						
FP16	3	6	0.8700	0.8700	0.9953	39.85

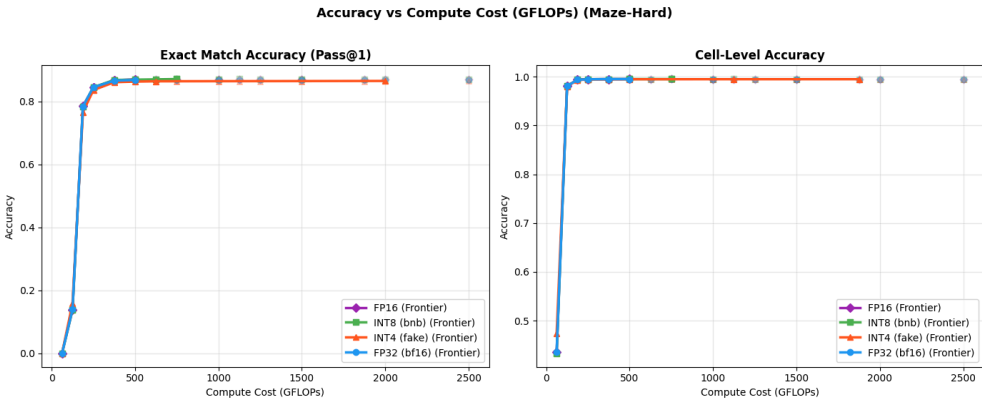
1125.0						
FP16	3	8	0.8700	0.8700	0.9953	52.92
1500.0						
FP16	3	10	0.8700	0.8700	0.9953	66.13
1875.0						
FP16	4	1	0.8440	0.8440	0.9949	9.24
250.0						
FP16	4	2	0.8700	0.8700	0.9953	18.00
500.0						
FP16	4	4	0.8700	0.8700	0.9953	35.52
1000.0						
FP16	4	6	0.8700	0.8700	0.9953	53.03
1500.0						
FP16	4	8	0.8700	0.8700	0.9953	70.54
2000.0						
FP16	4	10	0.8700	0.8700	0.9953	87.94
2500.1						
INT8 (bnb)	1	1	0.0000	0.0000	0.4333	1.86
62.5						
INT8 (bnb)	1	2	0.1380	0.1380	0.9808	3.38
125.0						
INT8 (bnb)	1	4	0.8450	0.8450	0.9949	6.35
250.0						
INT8 (bnb)	1	6	0.8680	0.8680	0.9953	9.21
375.0						
INT8 (bnb)	1	8	0.8690	0.8690	0.9953	12.05
500.0						
INT8 (bnb)	1	10	0.8700	0.8700	0.9953	14.90
625.0						
INT8 (bnb)	2	1	0.1380	0.1380	0.9808	3.37
125.0						
INT8 (bnb)	2	2	0.8450	0.8450	0.9949	6.33
250.0						
INT8 (bnb)	2	4	0.8690	0.8690	0.9953	12.02
500.0						
INT8 (bnb)	2	6	0.8710	0.8710	0.9953	17.70
750.0						
INT8 (bnb)	2	8	0.8700	0.8700	0.9953	23.38
1000.0						
INT8 (bnb)	2	10	0.8700	0.8700	0.9953	29.06
1250.0						
INT8 (bnb)	3	1	0.7830	0.7830	0.9940	4.88
187.5						
INT8 (bnb)	3	2	0.8680	0.8680	0.9953	9.17
375.0						
INT8 (bnb)	3	4	0.8710	0.8710	0.9953	17.68
750.0						
INT8 (bnb)	3	6	0.8710	0.8710	0.9953	26.20
1125.0						
INT8 (bnb)	3	8	0.8700	0.8700	0.9953	34.71
1500.0						

INT8 (bnb) 1875.0	3	10	0.8700	0.8700	0.9953	43.22
INT8 (bnb) 250.0	4	1	0.8450	0.8450	0.9949	6.32
INT8 (bnb) 500.0	4	2	0.8690	0.8690	0.9953	12.01
INT8 (bnb) 1000.0	4	4	0.8700	0.8700	0.9953	23.35
INT8 (bnb) 1500.0	4	6	0.8700	0.8700	0.9953	34.69
INT8 (bnb) 2000.0	4	8	0.8700	0.8700	0.9953	46.03
INT8 (bnb) 2500.1	4	10	0.8700	0.8700	0.9953	57.38
INT4 (fake) 62.5	1	1	0.0000	0.0000	0.4734	1.08
INT4 (fake) 125.0	1	2	0.1580	0.1580	0.9805	1.69
INT4 (fake) 250.0	1	4	0.8360	0.8360	0.9945	2.89
INT4 (fake) 375.0	1	6	0.8610	0.8610	0.9949	4.09
INT4 (fake) 500.0	1	8	0.8630	0.8630	0.9950	5.29
INT4 (fake) 625.0	1	10	0.8640	0.8640	0.9950	6.49
INT4 (fake) 125.0	2	1	0.1580	0.1580	0.9805	1.67
INT4 (fake) 250.0	2	2	0.8360	0.8360	0.9945	2.87
INT4 (fake) 500.0	2	4	0.8630	0.8630	0.9950	5.26
INT4 (fake) 750.0	2	6	0.8640	0.8640	0.9950	7.64
INT4 (fake) 1000.0	2	8	0.8630	0.8630	0.9950	10.03
INT4 (fake) 1250.0	2	10	0.8640	0.8640	0.9950	12.42
INT4 (fake) 187.5	3	1	0.7660	0.7660	0.9934	2.27
INT4 (fake) 375.0	3	2	0.8610	0.8610	0.9949	4.05
INT4 (fake) 750.0	3	4	0.8640	0.8640	0.9950	7.62
INT4 (fake) 1125.0	3	6	0.8630	0.8630	0.9950	11.20
INT4 (fake) 1500.0	3	8	0.8640	0.8640	0.9950	14.77
INT4 (fake) 1875.0	3	10	0.8640	0.8640	0.9950	18.34
INT4 (fake)	4	1	0.8360	0.8360	0.9945	2.86

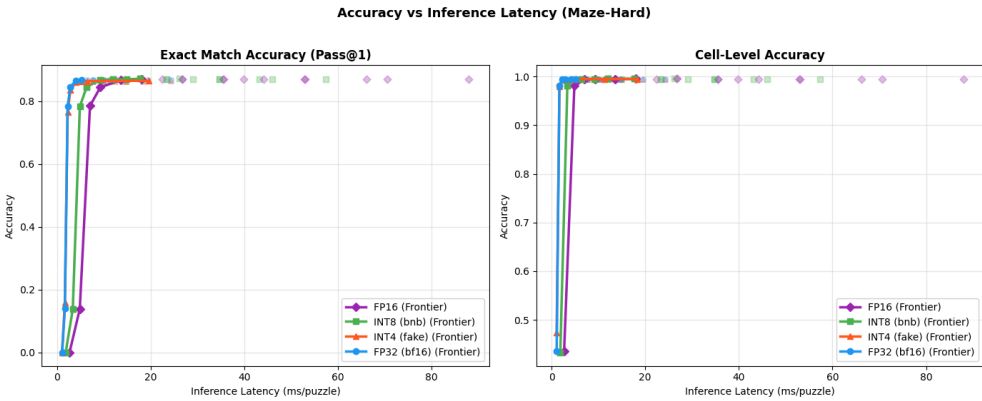
250.0						
INT4 (fake)	4	2	0.8630	0.8630	0.9950	5.24
500.0						
INT4 (fake)	4	4	0.8630	0.8630	0.9950	9.99
1000.0						
INT4 (fake)	4	6	0.8640	0.8640	0.9950	14.75
1500.0						
INT4 (fake)	4	8	0.8650	0.8650	0.9950	19.51
2000.0						
INT4 (fake)	4	10	0.8650	0.8650	0.9950	24.27
2500.1						
FP32 (bf16)	1	1	0.0000	0.0000	0.4361	1.08
62.5						
FP32 (bf16)	1	2	0.1390	0.1390	0.9808	1.68
125.0						
FP32 (bf16)	1	4	0.8450	0.8450	0.9949	2.88
250.0						
FP32 (bf16)	1	6	0.8660	0.8660	0.9952	4.08
375.0						
FP32 (bf16)	1	8	0.8680	0.8680	0.9953	5.28
500.0						
FP32 (bf16)	1	10	0.8680	0.8680	0.9952	6.48
625.0						
FP32 (bf16)	2	1	0.1390	0.1390	0.9808	1.67
125.0						
FP32 (bf16)	2	2	0.8450	0.8450	0.9949	2.87
250.0						
FP32 (bf16)	2	4	0.8680	0.8680	0.9953	5.25
500.0						
FP32 (bf16)	2	6	0.8680	0.8680	0.9952	7.64
750.0						
FP32 (bf16)	2	8	0.8680	0.8680	0.9952	10.02
1000.0						
FP32 (bf16)	2	10	0.8680	0.8680	0.9952	12.41
1250.0						
FP32 (bf16)	3	1	0.7840	0.7840	0.9940	2.27
187.5						
FP32 (bf16)	3	2	0.8660	0.8660	0.9952	4.05
375.0						
FP32 (bf16)	3	4	0.8680	0.8680	0.9952	7.62
750.0						
FP32 (bf16)	3	6	0.8680	0.8680	0.9952	11.19
1125.0						
FP32 (bf16)	3	8	0.8680	0.8680	0.9952	14.76
1500.0						
FP32 (bf16)	3	10	0.8680	0.8680	0.9952	18.33
1875.0						
FP32 (bf16)	4	1	0.8450	0.8450	0.9949	2.86
250.0						
FP32 (bf16)	4	2	0.8680	0.8680	0.9953	5.24
500.0						

FP32 (bf16)	4	4	0.8680	0.8680	0.9952	9.99
1000.0						
FP32 (bf16)	4	6	0.8680	0.8680	0.9952	14.74
1500.0						
FP32 (bf16)	4	8	0.8680	0.8680	0.9952	19.50
2000.0						
FP32 (bf16)	4	10	0.8680	0.8680	0.9952	24.25
2500.1						

6.2 Plot depth × quantization grid



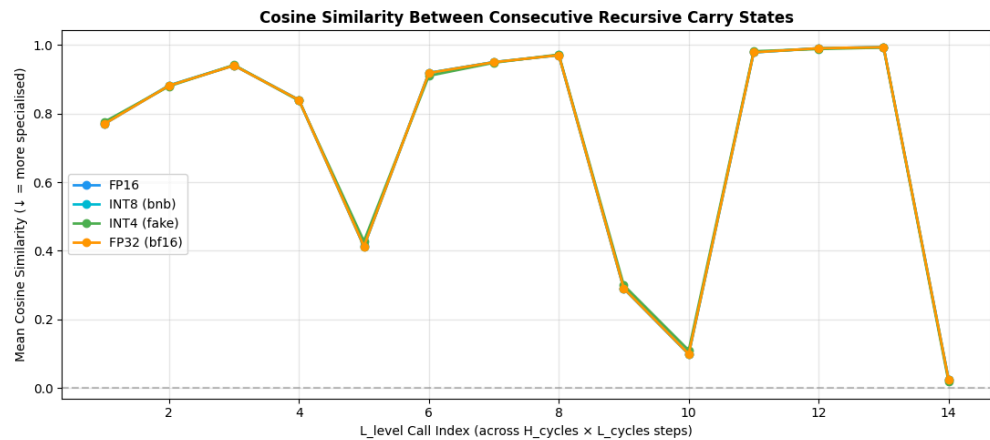
Plot from cell 34



Plot from cell 34

7.1 Carry-state similarity hooks

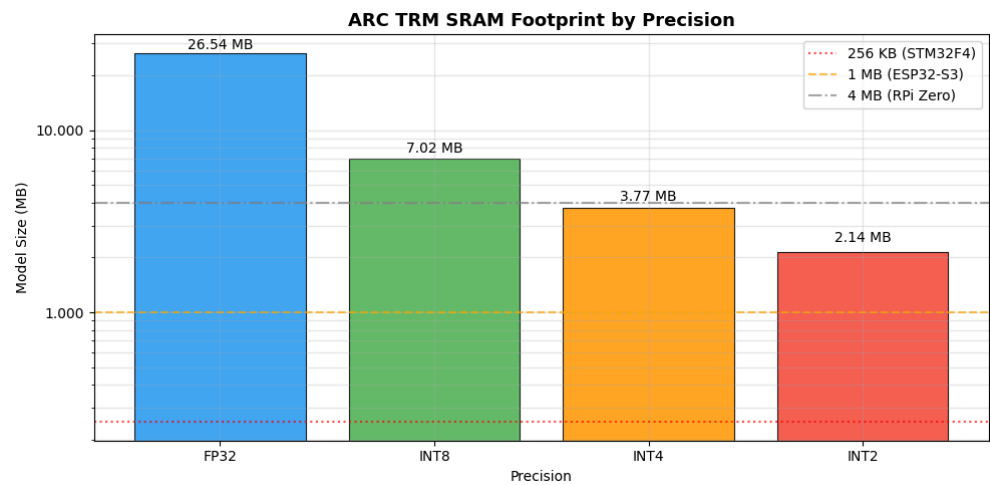
7.2 Compare carry similarity across quantization variants



Plot from cell 37

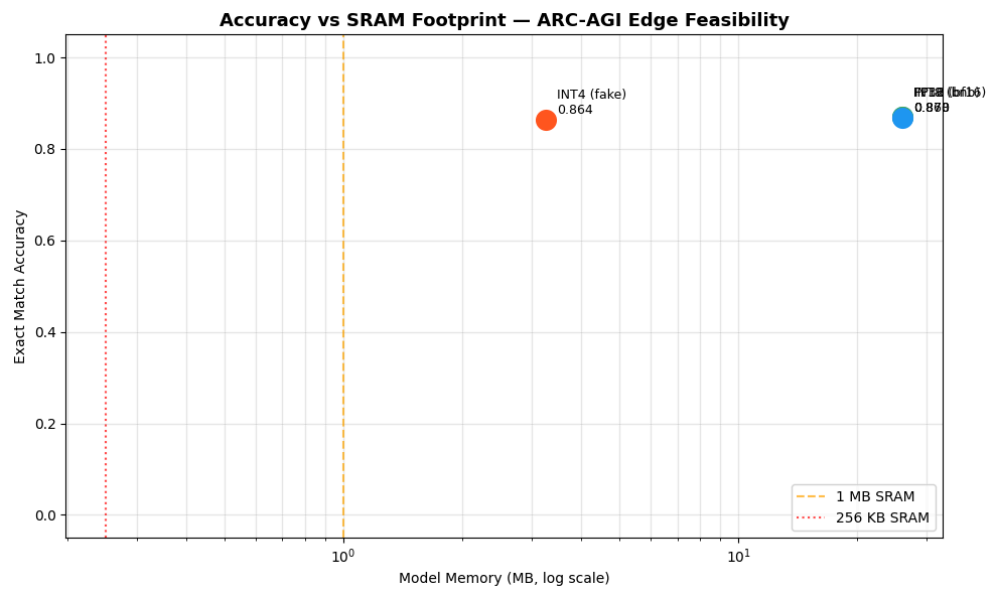
8.1 Comprehensive model footprint analysis

8.2 SRAM footprint visualisation



Plot from cell 41

8.3 Accuracy vs SRAM: the key publishable plot



Plot from cell 42

9.1 Inspect puzzle_emb shape & byte count

9.2 Option A — INT8 quantize the embedding buffer

9.3 Option B — SVD Low-Rank Puzzle Embedding

9.4 Option D — Single-Puzzle Inference (only load 1 row)

9.5 Footprint summary across embedding strategies

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

11.1 Per-channel calibration statistics

11.2 Apply calibrated INT4 and evaluate

12.1 QAT fine-tuning loop

13.1 Magnitude-based row pruning

14.1 TorchScript trace of single TRM backbone step

14.2 Latency benchmark: traced model vs eager

15.1 Build train / val loaders

15.2 QAT loop with val-set evaluation every 10 steps

15.3 Plot train loss + val accuracy curve

16.1 Build fused INT8 + SinglePuzzle model

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

16.3 Save fused model (state dict + embedding table)

17.1 CPU-only latency benchmark

17.2 Peak SRAM usage during inference (tracemalloc)

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 187498.4 MFLOPs
Full inference (10 steps) : 1874.98 GFLOPs

Power estimate (Cortex-M55 @ ~1 GFLOPS/s, ~1 mW/GFLOP):
FLOPs : 1874.984 GFLOPs/puzzle
Est. energy : ~1874.98 mJ/puzzle
Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

=== Final Deployment Scorecard ===
Backbone MB Total deploy MB Puzzle Exact Cell
Acc Fits 4MB Fits 8MB
Config

FP32 (baseline)			26.03	26.03	0.8680
0.9952	x	x			
INT8 (bnb)			6.51	6.51	0.8700
0.9953	x	✓			
INT4 (calibrated)			3.25	3.26	0.8640
0.9950	✓	✓			
INT8 + Single-Puzzle ★			6.51	6.51	1.0000
1.0000	x	✓			

★ = recommended target for 8MB SRAM devices

Utility: fix H_init / L_init device mismatch (permanent patch)

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

VARIANT: FP16		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def) 0.8700 0.9953 105.6 ms 3000.1	Exact	0.7850	0.8670	0.8700	0.8700
	Cell	0.9940	0.9953	0.9953	0.9953
	Latency	7.0 ms	13.6 ms	26.7 ms	53.0 ms
	GFLOPs	187.5	375.0	750.0	1500.0
1 0.8700 0.9953 35.6 ms 1000.0	Exact	0.0000	0.1380	0.8440	0.8700
	Cell	0.4362	0.9808	0.9949	0.9953
	Latency	2.7 ms	4.9 ms	9.3 ms	18.0 ms
	GFLOPs	62.5	125.0	250.0	500.0
VARIANT: INT8 (bnb)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					

3 (Def) 0.8700 0.9953 68.8 ms 3000.1	Exact	0.7830	0.8680	0.8710	0.8700
	Cell	0.9940	0.9953	0.9953	0.9953
	Latency	4.9 ms	9.2 ms	17.7 ms	34.7 ms
	GFLOPs	187.5	375.0	750.0	1500.0

1 0.8700 0.9953 23.5 ms 1000.0	Exact	0.0000	0.1380	0.8450	0.8690
	Cell	0.4333	0.9808	0.9949	0.9953
	Latency	1.9 ms	3.4 ms	6.3 ms	12.1 ms
	GFLOPs	62.5	125.0	250.0	500.0
=====					

VARIANT: INT4 (fake)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					

3 (Def) 0.8640 0.9950 29.1 ms 3000.1	Exact	0.7660	0.8610	0.8640	0.8640
	Cell	0.9934	0.9949	0.9950	0.9950
	Latency	2.3 ms	4.1 ms	7.6 ms	14.8 ms
	GFLOPs	187.5	375.0	750.0	1500.0

1 0.8630 0.9950 10.1 ms 1000.0	Exact	0.0000	0.1580	0.8360	0.8630
	Cell	0.4734	0.9805	0.9945	0.9950
	Latency	1.1 ms	1.7 ms	2.9 ms	5.3 ms
	GFLOPs	62.5	125.0	250.0	500.0
=====					

=====					
VARIANT: FP32 (bf16)			(device: cuda)		
=====					

H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					

3 (Def)	Exact	0.7850	0.8670	0.8700	0.8700
0.8700	Cell	0.9940	0.9953	0.9953	0.9953
0.9953	Latency	7.1 ms	13.7 ms	26.9 ms	53.3 ms
105.9 ms	GFLOPs	187.5	375.0	750.0	1500.0
3000.1					

1	Exact	0.0000	0.1390	0.8440	0.8690
0.8700	Cell	0.4363	0.9808	0.9949	0.9953
0.9953	Latency	2.7 ms	4.9 ms	9.3 ms	18.1 ms
35.8 ms	GFLOPs	62.5	125.0	250.0	500.0
1000.0					
=====					

19.1 Zero-pad truncation at various context lengths

Full seq_len = 900 (FLOPs $\propto$ seq_len²)

Context	% of full	Attn FLOPs	Puzzle Exact	Cell Acc

900	100%	~187 GFLOPs	0.7850	0.9940
450	50%	~47 GFLOPs	0.0000	0.6576
225	25%	~12 GFLOPs	0.0000	0.5038
128	14%	~4 GFLOPs	0.0000	0.4436

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

21.1 Instantiate student TRM and train with KL distillation

Teacher params : 6,822,914
Student params : 855,554 (0.13× teacher)
Resuming training from checkpoint:
checkpoints/student_distill_latest.pt
Resumed at step 600
Step 640/1000 | kl=2976.6737 | hard=0.3435
Step 680/1000 | kl=2909.6919 | hard=0.3440


```
04/06/2026, 21:25 EdgeTRM Model Evaluation & Compression Report
Step 720/1000 | kl=2807.1469 | hard=0.3739
Step 760/1000 | kl=2626.5060 | hard=0.3488
Step 800/1000 | kl=2619.8455 | hard=0.3641
[Checkpoint] Saved checkpoint at step 800 to
checkpoints/student_distill_latest.pt
Step 840/1000 | kl=2377.3866 | hard=0.3061
Step 880/1000 | kl=2502.1829 | hard=0.4040
Step 920/1000 | kl=2261.5046 | hard=0.2868
Step 960/1000 | kl=2238.7907 | hard=0.3175
Step 1000/1000 | kl=2064.3248 | hard=0.2746
[Checkpoint] Saved checkpoint at step 1000 to
checkpoints/student_distill_latest.pt
Evaluating student on test set...
Student (hidden=256, 1L, 1-cycle): exact=0.0000 cell=0.8725 ~47
GFL0Ps
```

22.1 Monkey-patch softmax attention → linear attention

Patched 2 Attention modules → linear O(n) attention
Theoretical compute reduction: ~14× in attention layers

Evaluating linear attention model on test set (n_sup_max=1)...

Evaluating softmax attention (FP32 baseline, n_sup_max=1) for comparison...

Model	Puzzle Exact	Cell Acc

FP32 softmax attn	0.7850	0.9940
Linear attn (ELU)	0.0000	0.8728

Accuracy cost of linear swap: +12.12pp cell accuracy
Note: linear attn needs retraining to recover accuracy – this is a zero-shot test.

22.2 Retrain linear attention model to recover accuracy

Sudoku Extreme Evaluation Outputs

MUST RUN FIRST — fix duplicate trm.py on Modal volume

Cell 9: Sparsity Audit

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

4.1 Sudoku Extreme DataLoader from pre-built .npz files

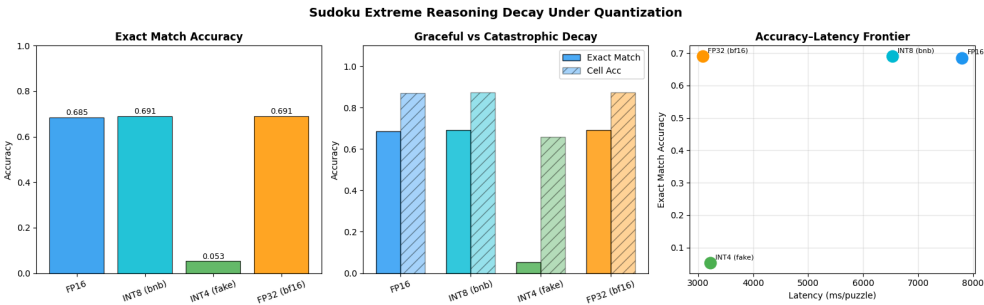
4.3 Baseline: evaluate the loaded FP32 model

4.4 Quantization helpers

4.7 Instantiate all model variants

5.1 Evaluate all quantization levels

5.2 Visualise reasoning decay



Plot from cell 31

6.1 Recursive depth sweep at inference time

Quant GFLOPs	H	n_sup	Pass@1	Pass@2	Cell Acc	Latency

FP16 8.5	1	1	0.0060	0.0060	0.5526	0.31
FP16 17.1	1	2	0.0210	0.0210	0.6941	0.57
FP16 34.2	1	4	0.3830	0.3830	0.7856	1.08
FP16 51.3	1	6	0.4960	0.4960	0.8151	1.59
FP16 68.3	1	8	0.5480	0.5480	0.8303	2.10
FP16 85.4	1	10	0.5810	0.5810	0.8389	2.62
FP16 17.1	2	1	0.0210	0.0210	0.6941	0.56
FP16 34.2	2	2	0.3830	0.3830	0.7856	1.08
FP16 68.3	2	4	0.5480	0.5480	0.8303	2.10
FP16 102.5	2	6	0.6100	0.6100	0.8488	3.12
FP16 136.7	2	8	0.6390	0.6390	0.8574	4.14
FP16 170.9	2	10	0.6580	0.6580	0.8637	5.16
FP16 25.6	3	1	0.2830	0.2830	0.7639	0.82
FP16 51.3	3	2	0.4960	0.4960	0.8151	1.58
FP16 102.5	3	4	0.6100	0.6100	0.8488	3.12
FP16	3	6	0.6490	0.6490	0.8601	4.65

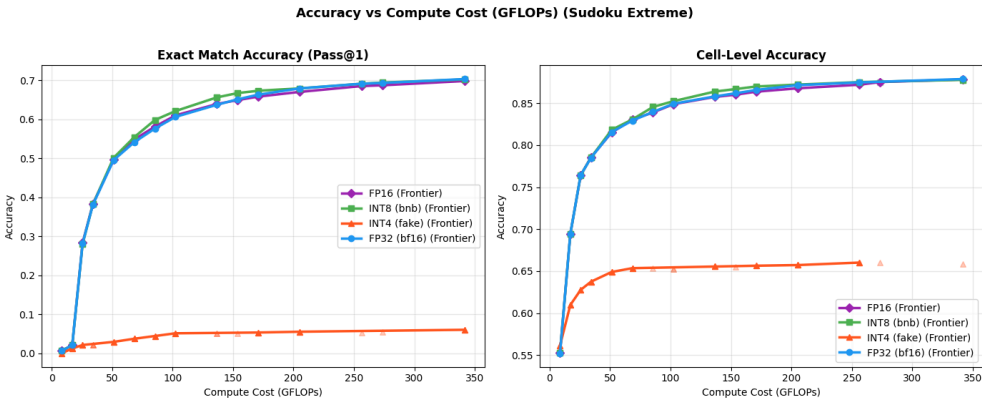
153.8						
FP16	3	8	0.6700	0.6700	0.8677	6.18
205.0						
FP16	3	10	0.6850	0.6850	0.8720	7.71
256.3						
FP16	4	1	0.3830	0.3830	0.7856	1.07
34.2						
FP16	4	2	0.5480	0.5480	0.8303	2.09
68.3						
FP16	4	4	0.6390	0.6390	0.8574	4.14
136.7						
FP16	4	6	0.6700	0.6700	0.8677	6.18
205.0						
FP16	4	8	0.6870	0.6870	0.8751	8.22
273.4						
FP16	4	10	0.6980	0.6980	0.8782	10.26
341.7						
INT8 (bnb)	1	1	0.0060	0.0060	0.5528	0.26
8.5						
INT8 (bnb)	1	2	0.0210	0.0210	0.6942	0.47
17.1						
INT8 (bnb)	1	4	0.3820	0.3820	0.7852	0.90
34.2						
INT8 (bnb)	1	6	0.5010	0.5010	0.8183	1.33
51.3						
INT8 (bnb)	1	8	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	1	10	0.5980	0.5980	0.8455	2.18
85.4						
INT8 (bnb)	2	1	0.0210	0.0210	0.6942	0.47
17.1						
INT8 (bnb)	2	2	0.3820	0.3820	0.7852	0.90
34.2						
INT8 (bnb)	2	4	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	2	6	0.6210	0.6210	0.8524	2.60
102.5						
INT8 (bnb)	2	8	0.6560	0.6560	0.8638	3.44
136.7						
INT8 (bnb)	2	10	0.6730	0.6730	0.8698	4.29
170.9						
INT8 (bnb)	3	1	0.2800	0.2800	0.7643	0.68
25.6						
INT8 (bnb)	3	2	0.5010	0.5010	0.8183	1.32
51.3						
INT8 (bnb)	3	4	0.6210	0.6210	0.8524	2.59
102.5						
INT8 (bnb)	3	6	0.6670	0.6670	0.8668	3.86
153.8						
INT8 (bnb)	3	8	0.6790	0.6790	0.8721	5.13
205.0						

INT8 (bnb)	3	10	0.6910	0.6910	0.8751	6.41
256.3						
INT8 (bnb)	4	1	0.3820	0.3820	0.7852	0.89
34.2						
INT8 (bnb)	4	2	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	4	4	0.6560	0.6560	0.8638	3.44
136.7						
INT8 (bnb)	4	6	0.6790	0.6790	0.8721	5.12
205.0						
INT8 (bnb)	4	8	0.6940	0.6940	0.8750	6.81
273.4						
INT8 (bnb)	4	10	0.7020	0.7020	0.8778	8.50
341.7						
INT4 (fake)	1	1	0.0000	0.0000	0.5612	0.16
8.5						
INT4 (fake)	1	2	0.0120	0.0120	0.6097	0.26
17.1						
INT4 (fake)	1	4	0.0210	0.0210	0.6375	0.48
34.2						
INT4 (fake)	1	6	0.0290	0.0290	0.6490	0.68
51.3						
INT4 (fake)	1	8	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	1	10	0.0440	0.0440	0.6530	1.11
85.4						
INT4 (fake)	2	1	0.0120	0.0120	0.6097	0.26
17.1						
INT4 (fake)	2	2	0.0210	0.0210	0.6375	0.47
34.2						
INT4 (fake)	2	4	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	2	6	0.0510	0.0510	0.6524	1.31
102.5						
INT4 (fake)	2	8	0.0510	0.0510	0.6555	1.73
136.7						
INT4 (fake)	2	10	0.0530	0.0530	0.6565	2.15
170.9						
INT4 (fake)	3	1	0.0210	0.0210	0.6277	0.37
25.6						
INT4 (fake)	3	2	0.0290	0.0290	0.6490	0.68
51.3						
INT4 (fake)	3	4	0.0510	0.0510	0.6524	1.31
102.5						
INT4 (fake)	3	6	0.0510	0.0510	0.6553	1.93
153.8						
INT4 (fake)	3	8	0.0550	0.0550	0.6573	2.56
205.0						
INT4 (fake)	3	10	0.0530	0.0530	0.6602	3.19
256.3						
INT4 (fake)	4	1	0.0210	0.0210	0.6375	0.47

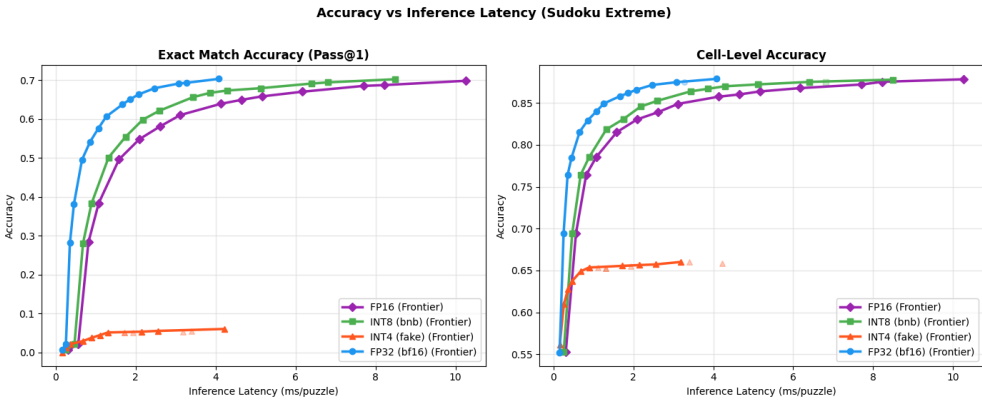
34.2						
INT4 (fake)	4	2	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	4	4	0.0510	0.0510	0.6555	1.72
136.7						
INT4 (fake)	4	6	0.0550	0.0550	0.6573	2.56
205.0						
INT4 (fake)	4	8	0.0550	0.0550	0.6601	3.40
273.4						
INT4 (fake)	4	10	0.0600	0.0600	0.6588	4.23
341.7						
FP32 (bf16)	1	1	0.0060	0.0060	0.5524	0.16
8.5						
FP32 (bf16)	1	2	0.0210	0.0210	0.6943	0.26
17.1						
FP32 (bf16)	1	4	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	1	6	0.4950	0.4950	0.8156	0.66
51.3						
FP32 (bf16)	1	8	0.5410	0.5410	0.8292	0.87
68.3						
FP32 (bf16)	1	10	0.5760	0.5760	0.8396	1.07
85.4						
FP32 (bf16)	2	1	0.0210	0.0210	0.6943	0.25
17.1						
FP32 (bf16)	2	2	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	2	4	0.5410	0.5410	0.8292	0.86
68.3						
FP32 (bf16)	2	6	0.6060	0.6060	0.8493	1.27
102.5						
FP32 (bf16)	2	8	0.6370	0.6370	0.8581	1.67
136.7						
FP32 (bf16)	2	10	0.6630	0.6630	0.8657	2.08
170.9						
FP32 (bf16)	3	1	0.2810	0.2810	0.7642	0.36
25.6						
FP32 (bf16)	3	2	0.4950	0.4950	0.8156	0.66
51.3						
FP32 (bf16)	3	4	0.6060	0.6060	0.8493	1.26
102.5						
FP32 (bf16)	3	6	0.6510	0.6510	0.8618	1.87
153.8						
FP32 (bf16)	3	8	0.6790	0.6790	0.8714	2.47
205.0						
FP32 (bf16)	3	10	0.6910	0.6910	0.8747	3.08
256.3						
FP32 (bf16)	4	1	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	4	2	0.5410	0.5410	0.8292	0.86
68.3						

FP32 (bf16)	4	4	0.6370	0.6370	0.8581	1.67
136.7						
FP32 (bf16)	4	6	0.6790	0.6790	0.8714	2.47
205.0						
FP32 (bf16)	4	8	0.6930	0.6930	0.8745	3.28
273.4						
FP32 (bf16)	4	10	0.7030	0.7030	0.8787	4.08
341.7						

6.2 Plot depth × quantization grid



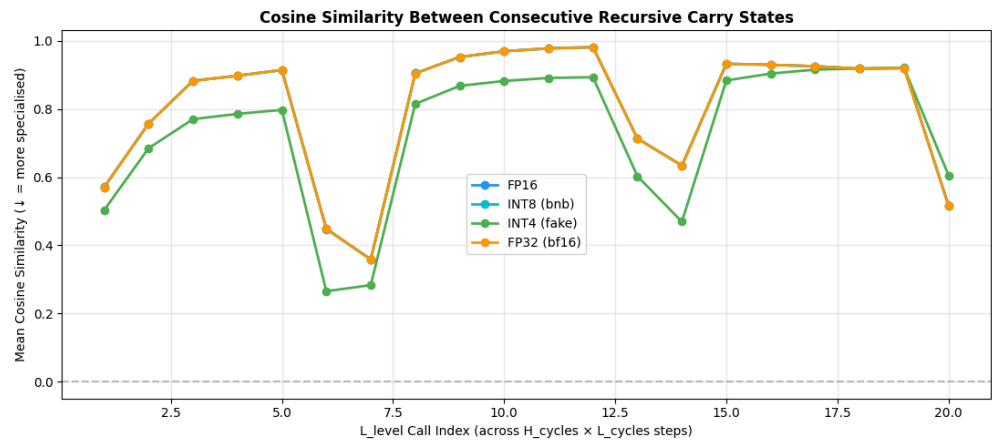
Plot from cell 34



Plot from cell 34

7.1 Carry-state similarity hooks

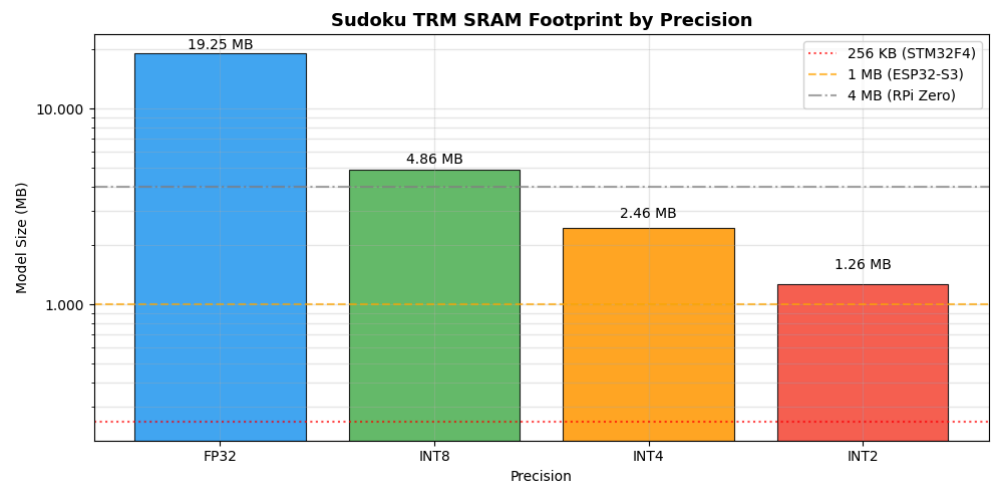
7.2 Compare carry similarity across quantization variants



Plot from cell 37

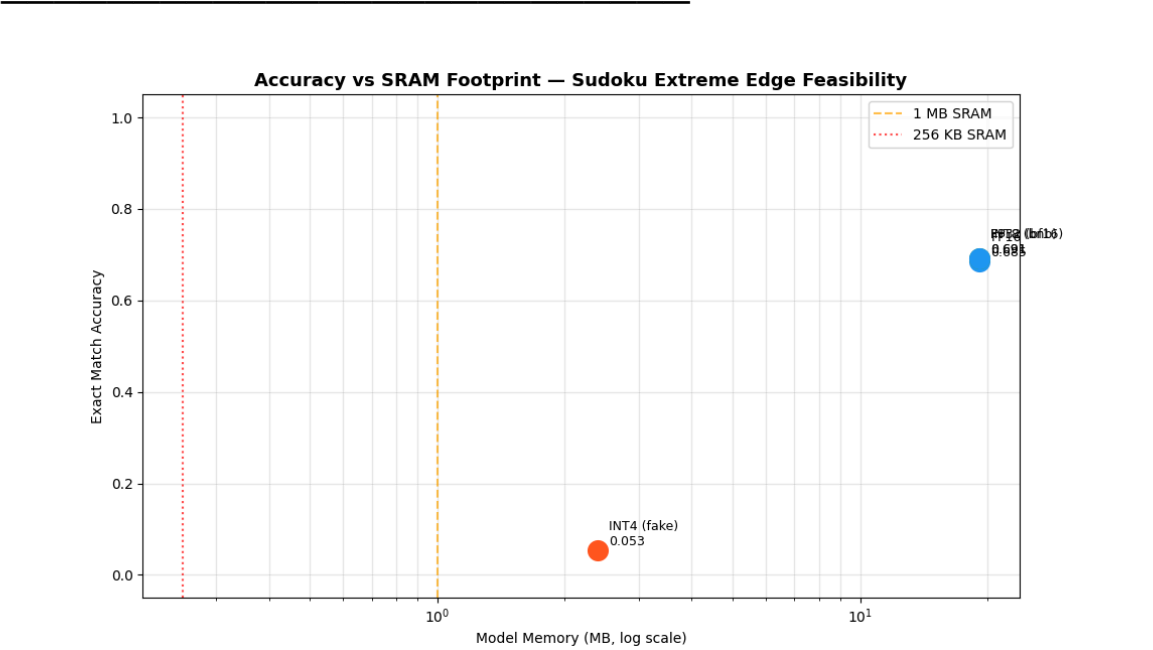
8.1 Comprehensive model footprint analysis

8.2 SRAM footprint visualisation



Plot from cell 41

8.3 Accuracy vs SRAM: the key publishable plot



Plot from cell 42

9.1 Inspect puzzle_emb shape & byte count

9.2 Option A — INT8 quantize the embedding buffer

9.3 Option B — SVD Low-Rank Puzzle Embedding

9.4 Option D — Single-Puzzle Inference (only load 1 row)

9.5 Footprint summary across embedding strategies

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

11.1 Per-channel calibration statistics

11.2 Apply calibrated INT4 and evaluate

12.1 QAT fine-tuning loop

13.1 Magnitude-based row pruning

14.1 TorchScript trace of single TRM backbone step

14.2 Latency benchmark: traced model vs eager

15.1 Build train / val loaders

15.2 QAT loop with val-set evaluation every 10 steps

15.3 Plot train loss + val accuracy curve

16.1 Build fused INT8 + SinglePuzzle model

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

16.3 Save fused model (state dict + embedding table)

17.1 CPU-only latency benchmark

17.2 Peak SRAM usage during inference (tracemalloc)

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 25660.1 MFLOPs
Full inference (10 steps) : 256.60 GFLOPs

Power estimate (Cortex-M55 @ ~1 GFLOPS/s, ~1 mW/GFLOP):
FLOPs : 256.601 GFLOPs/puzzle
Est. energy : ~256.60 mJ/puzzle
Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

=== Final Deployment Scorecard ===
Backbone MB Total deploy MB Puzzle Exact Cell
Acc Fits 4MB Fits 8MB
Config

FP32 (baseline)			19.18	19.19	0.6910
0.8747	x	x			
INT8 (bnb)			4.80	4.80	0.6910
0.8751	x	✓			
INT4 (calibrated)			2.40	2.40	0.0530
0.6602	✓	✓			
INT8 + Single-Puzzle ★			4.80	4.80	0.8600
0.9464	x	✓			

★ = recommended target for 8MB SRAM devices

Utility: fix H_init / L_init device mismatch (permanent patch)

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

VARIANT: FP16		(device: cuda)			
H_cycles n=16	Metric	n=1	n=2	n=4	n=8
3 (Def)	Exact	0.2830	0.4960	0.6100	0.6700
0.7160	Cell	0.7639	0.8151	0.8488	0.8677
0.8851	Latency	0.8 ms	1.6 ms	3.1 ms	6.2 ms
12.3 ms	GFLOPs	25.6	51.3	102.5	205.0
410.1					
1	Exact	0.0060	0.0210	0.3830	0.5480
0.6390	Cell	0.5526	0.6941	0.7856	0.8303
0.8574	Latency	0.3 ms	0.6 ms	1.1 ms	2.1 ms
4.2 ms	GFLOPs	8.5	17.1	34.2	68.4
136.7					
VARIANT: INT8 (bnb)		(device: cuda)			
H_cycles n=16	Metric	n=1	n=2	n=4	n=8

3 (Def) 0.7120 0.8816 10.2 ms 410.1	Exact	0.2800	0.5010	0.6210	0.6790
	Cell	0.7643	0.8183	0.8524	0.8721
	Latency	0.7 ms	1.3 ms	2.6 ms	5.2 ms
	GFLOPs	25.6	51.3	102.5	205.0

1 0.6560 0.8638 3.5 ms 136.7	Exact	0.0060	0.0210	0.3820	0.5540
	Cell	0.5528	0.6942	0.7852	0.8306
	Latency	0.3 ms	0.5 ms	0.9 ms	1.8 ms
	GFLOPs	8.5	17.1	34.2	68.4
=====					

VARIANT: INT4 (fake) (device: cuda)

H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def) 0.0610 0.6597 5.1 ms 410.1	Exact	0.0210	0.0290	0.0510	0.0550
	Cell	0.6277	0.6490	0.6524	0.6573
	Latency	0.4 ms	0.7 ms	1.3 ms	2.6 ms
	GFLOPs	25.6	51.3	102.5	205.0

1 0.0510 0.6555 1.7 ms 136.7	Exact	0.0000	0.0120	0.0210	0.0370
	Cell	0.5612	0.6097	0.6375	0.6536
	Latency	0.2 ms	0.3 ms	0.5 ms	0.9 ms
	GFLOPs	8.5	17.1	34.2	68.4
=====					

VARIANT: FP32 (bf16) (device: cuda)

H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					

3 (Def)	Exact	0.2840	0.4960	0.6040	0.6690
0.7090					
	Cell	0.7640	0.8147	0.8474	0.8680
0.8810					
	Latency	0.8 ms	1.6 ms	3.2 ms	6.3 ms
12.7 ms					
	GFLOPs	25.6	51.3	102.5	205.0
410.1					

1	Exact	0.0060	0.0210	0.3830	0.5470
0.6370					
	Cell	0.5527	0.6943	0.7856	0.8293
0.8563					
	Latency	0.3 ms	0.6 ms	1.1 ms	2.2 ms
4.3 ms					
	GFLOPs	8.5	17.1	34.2	68.4
136.7					
=====					

19.1 Zero-pad truncation at various context lengths

Full seq_len = 81 (FLOPs $\propto$ seq_len due to MLP architecture)

Context	% of full	MLP FLOPs	Puzzle Exact	Cell Acc

81	100%	~25.63 GFLOPs	0.2840	0.7640
40	49%	~14.80 GFLOPs	0.0000	0.3276
20	25%	~9.51 GFLOPs	0.0000	0.2035
16	20%	~8.46 GFLOPs	0.0000	0.1767

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

21.1 Instantiate student TRM and train with KL distillation

Teacher params : 5,028,866
Student params : 744,962 (0.15 $\times$ teacher)
Step 40/100 | kl=1802.0414 | hard=2.1362
Step 80/100 | kl=1125.4269 | hard=2.1942
Evaluating student on test set...
Student (hidden=256, 1L, 1-cycle): exact=0.0000 cell=0.5477 ~1 GFLOPs

22.1 Monkey-patch softmax attention → linear attention

22.2 Retrain linear attention model to recover accuracy
